

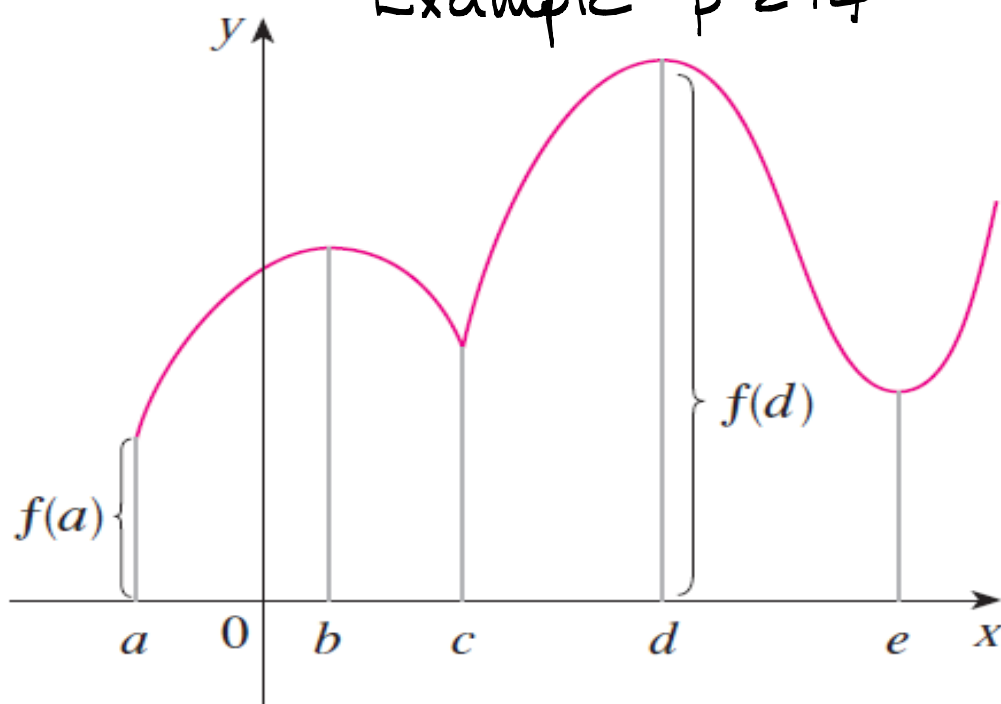
Theme 4: Applications of differentiation.

Note Title

2014/04/03

Unit 4.1 Maximum and minimum values pp. 274 - 282

Example p 274



Definition 1: p 274

Let c be a number in the domain D of a function. Then $f(c)$ is the

- absolute maximum of f on D if
$$f(c) \geq f(x) \quad \forall x \in D.$$
- absolute minimum of f on D if
$$f(c) \leq f(x) \quad \forall x \in D$$

Absolute/global extreme values.

Definition 2 The number $f(c)$ is a

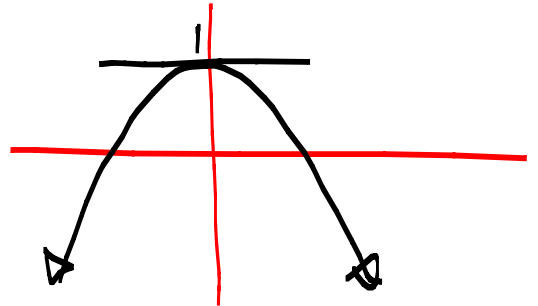
- local maximum if $f(c) \geq f(x)$ when x is near c .
- local minimum if $f(c) \leq f(x)$ when x is near c .

Local max/min — Local extremes.

Find the extremes of $f(x) = -x^2 + 1$

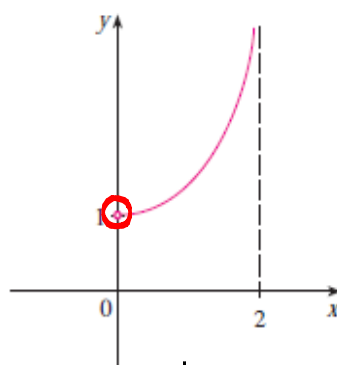
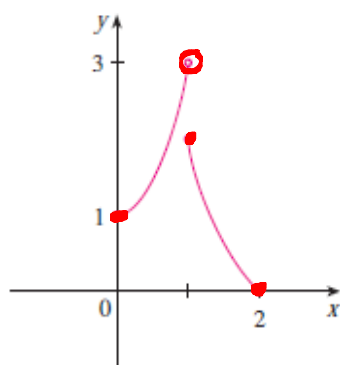
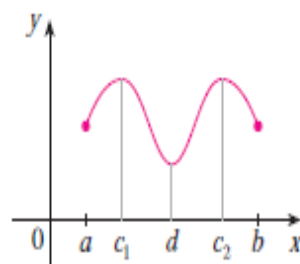
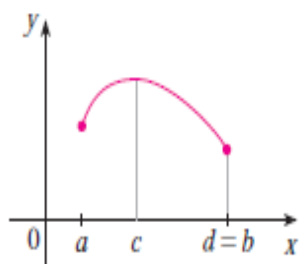
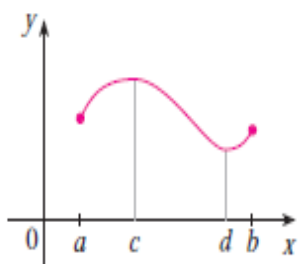
f has a
absolute max at
 $x = 0$ namely, 1 .

f has an abs. max $f(0) = 1$



Extreme value theorem p275

f continuous on closed interval $[a, b]$ then f attains an absolute max value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

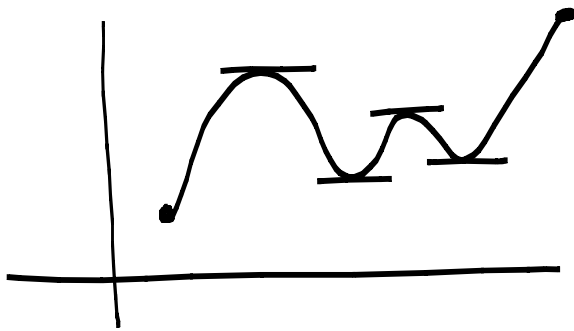


Abs min $f(2) = 0$, no abs max

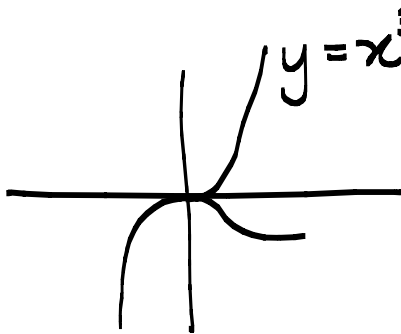
No max or min values

Fermat's theorem: p276

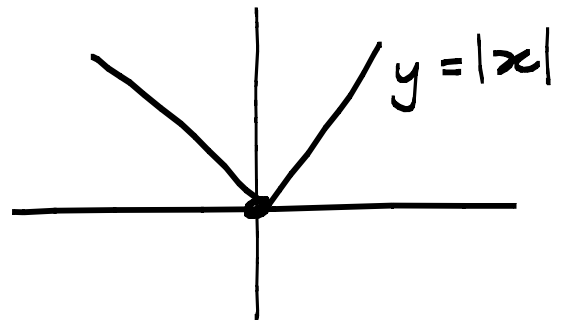
If f has a local max or min at c , and if $f'(c)$ exists, then $f'(c) = 0$.



Be careful: The converse the Fermat's theorem is not true.



$f'(0) = 0$
no local max/min



$f'(0)$ does not exist.
yet there is a local min

Definition 6 p278

A critical number of f is a number c in the domain of f such that

$f'(c) = 0$ or $f'(c)$ does not exist.

Example Also see example p278

① Find the critical values of

$$f(x) = (x-1) \cdot x^{2/3}$$

$$f'(x) = 1 \cdot x^{2/3} + (x-1) \frac{2}{3} x^{-1/3} = 0$$

$$\text{Solve } x^{2/3} + \frac{2}{3} x^{-1/3} (x-1) = 0$$

$$x^{2/3} + \frac{2(x-1)}{3x^{1/3}} = 0$$

$$\times 3x^{1/3}$$

$$3x + 2x - 2 = 0$$

$$\Rightarrow 5x = 2$$

$$x = \frac{2}{5}$$

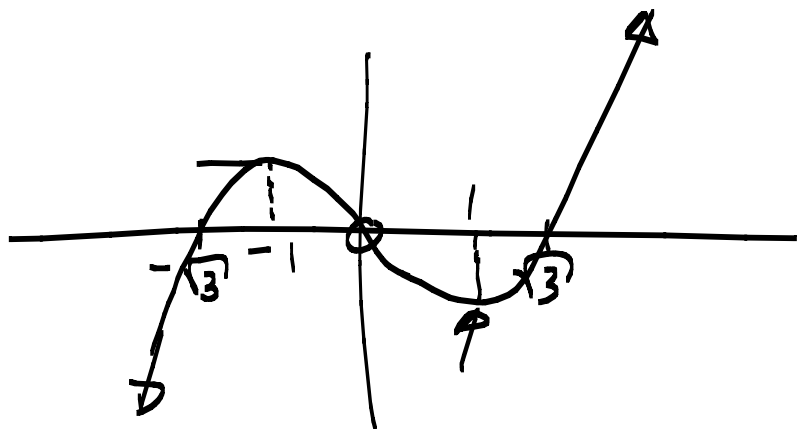
Critical value at $x = \frac{2}{5}$ and
at $x = 0$ (discontinuous at $x=0 \Rightarrow f'(0)$ does not exist)

② Find the critical values of the function and use a graph to decide whether the function has extreme values.

$$1. f(x) = x^3 - 3x$$

$$f'(x) = 3x^2 - 3 = 0$$

$\Rightarrow$ C.U's at $x = 1, -1$. Further $f'(x)$ is defined for all x .



local max at $x = -1$ of

$$f(-1) = (-1)^3 - 3(-1) = 2$$

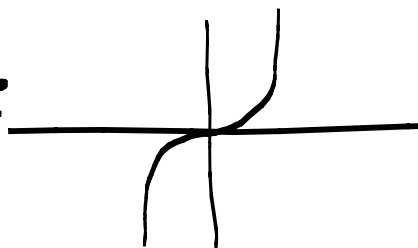
local min at $x = 1$ of

$$f(1) = -2$$

$$2. f(x) = x^3$$

$$f'(x) = 3x^2 = 0 \Rightarrow \text{CV's at } x=0$$

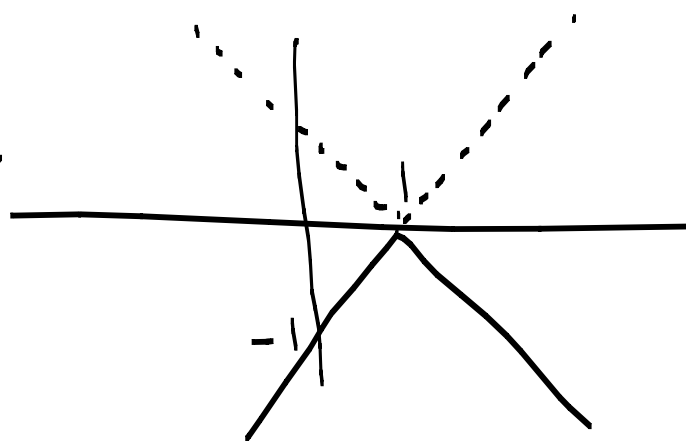
No local or absolute
extremes.



$$3. f(x) = -|x-1|$$

$f'(1)$ does not exist.

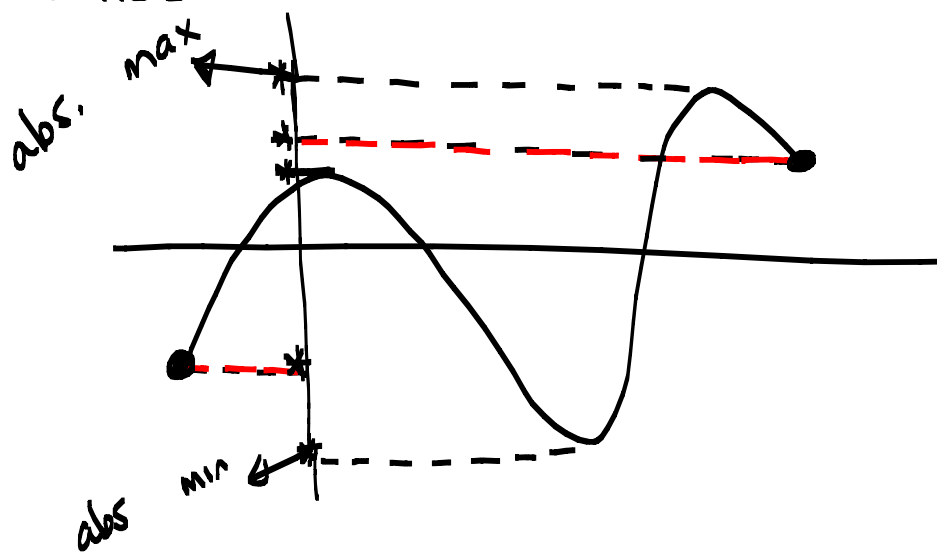
Absolute max at $x=1$
namely $f(1) = 0$



To find an absolute max or min of a continuous function on a closed interval:

The closed interval method p 278

1. Find the critical values of f in (a, b)
2. Find the function values of f at critical points and endpoints.
3. Largest - absolute maximum.
Smallest - absolute minimum.



Example: Find the extremes of

1. $f(x) = x^3 - 3x^2 + 1$, $x \in [-\frac{1}{2}, 4]$

① $f'(x) = 3x^2 - 6x$

$f'(x)$ exists everywhere

$$3x^2 - 6x = 0 \Rightarrow x = 0 \text{ or } x = 2$$

$$\Rightarrow \text{CV's } x = 0 \text{ or } x = 2$$

$$\begin{array}{lcl}
 \textcircled{2} & \begin{array}{l} f(-\frac{1}{2}) = \frac{1}{8} \\ f(0) = 1 \\ f(2) = -3 \\ f(4) = 17 \end{array} & \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{CV's} \\ \text{Endpoints} \end{array}
 \end{array}$$

- $\textcircled{3}$ $f(2) = -3$ is the absolute minimum at $x = 2$
 $f(4) = 17$ is the absolute maximum at $x = 4$.

2. Find the extremes of $f(x) = x^4 - 2x^2$ on $[-0.5, 2]$.

$$\textcircled{1} \quad f'(x) = 4x^3 - 4x$$

$f'(x)$ exists everywhere and

$$x^3 - x = 0 \Rightarrow x = 0, 1, -1.$$

CV's on $[-0.5, 2]$ is $x = 0, 1$.

$$\textcircled{2} \quad f(-0.5) = -0.4375$$

$$f(0) = 0$$

$$f(1) = -1$$

$$f(2) = 8$$

- $\textcircled{3}$ Abs max of 8 at $x = 2$
 Abs min of -1 at $x = 1$

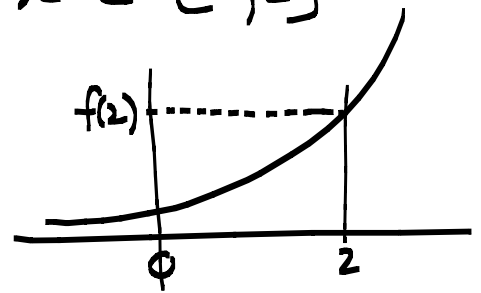
3. Find the max value of

$$f(x) = e^x \quad \text{on } x \in [0, 2]$$

$$f'(x) = e^x > 0 \quad \forall x \in [0, 2]$$

$\Rightarrow f$ increasing on $[0, 2]$

$\Rightarrow$ Absolute max of f
on $[0, 2]$ is $f(2) = e^2$



4. $f(x) = -x^2 + 1$ on $[\frac{1}{2}, 3]$

5. Find the loc extremes of

$$f(x) = |x^3 - 9x|$$